

Optical Coherence Tomography Techniques for Contextualizing and Reconstructing
Displacement Responses in the Mammalian Cochlea

Brian L. Frost

Submitted in partial fulfillment of the
requirements for the degree of
Doctor of Philosophy
under the Executive Committee
of the Graduate School of Arts and Sciences

COLUMBIA UNIVERSITY

2024

© 2024

Brian L. Frost

All Rights Reserved

Abstract

Optical Coherence Tomography Techniques for Contextualizing and Reconstructing Displacement Responses in the Mammalian Cochlea

Brian L. Frost

Spectral domain optical coherence tomography (OCT) is a powerful tool for measuring nanometer-scale displacement responses in the cochlea, as it is capable of volumetric imaging and vibrometry at a depth into a sample. The past decade has seen a wealth of OCT-measured displacement data from structures within the organ of Corti complex (OCC) that had previously been impossible to measure *in vivo*. These data have revealed surprising features of active intra-OCC motion but have not yet led to a complete understanding of cochlear amplification, the means by which active processes enhance the tuning and gain of the cochlear displacement responses in a level-dependent manner.

Certain technical challenges arise from the properties of OCT imaging and vibrometry that obscure the interpretation of intra-OCC displacement measurements. In particular, OCT-measured responses are dependent on the orientation of the system's beam axis. The beam axis is generally chosen based on experimental convenience, and has no inherent relevance to the anatomy of the cochlea.

This introduces two problems: 1) OCT-acquired images of the cochlea may be taken at skewed angles relative to the cochlea's naturally endowed anatomical coordinates, and 2) OCT-measured displacement responses are projections of a three-dimensional motion onto the beam axis. This thesis concerns the quantification of these effects on intra-OCC displacement measurements, as

well as the development of methods to overcome these complications *in vivo*. In doing so, previously reported data that appear to disagree can be synthesized.

I present a method by which the skew of OCT images relative to cochlear anatomy can be quantified, relating the OCT system's optical coordinates to the cochlear anatomy. With this method, I have shown that OCT images resembling familiar anatomical drawings of longitudinal cross-sections often capture a completely different anatomical slice of the cochlea. This leads to large quantitative shifts in phase responses when measuring displacements along a single beam axis, as opposed to what one would measure if s/he were measuring along an anatomically relevant axis. I have also provided a method by which to account for this phenomenon to capture structures related in some desired anatomical fashion.

I then turn to the issue of projection of the three-dimensional cochlear motion onto the OCT beam axis. I have provided a method for reconstructing two- and three-dimensional displacement responses in the relevant anatomical directions by acquiring displacement measurements at multiple locations within the cochlea. In doing so, I have revealed that previously unexplained disagreements between measurements in different experimental preparations can be explained by competing components of motion being projected onto the single axis. I have also shown that motion at the junction between the outer hair cells and Deiters cells follows a lineal pattern, as opposed to non-degenerate elliptical patterns that would be expected of fluid motion in this region. This method requires the acquisition of data at many points within the OCC, making it significantly time-consuming. This makes it vulnerable to sample drift and deterioration, and reduces experimental yield. Certain applications of the method – such as reconstructing displacement maps over a dense volume – are thereby intractable. To address this problem, I have developed a compressed sensing method for vibrometry (CSVi). CSVi is a classical optimization method based on a total generalized variation signal prior, which is shown to out-perform methods using total variation and wavelet domain sparsity priors. I have also found that uniform sub-sampling schema offered significant performance benefits over random sub-sampling schema. I found that this CSVi method can reconstruct densely sampled displacement maps in the

cochlea *in vivo* with less than 5% normalized mean square error, using only 10% of samples.

While these methods offer new insight into interpretation of OCT displacement measurements, there is still a challenge in measuring the motion of the stereocilia of the hair cells. The stereocilia are too small to be imaged using OCT, and the proxy measurement of differential motion of the reticular lamina and tectorial membrane (between which the stereocilia lie) is not yet achievable in the gerbil base. Stereocilia motion is related to the transduction current through the hair cells, which is critical to understanding cochlear function. These currents lead to neurotransmitter release and active electromotile responses believed to be responsible for cochlear amplification.

I present a model for studying another proxy measurement of the stereocilia motion – the voltage in the cochlea's scala tympani, or cochlear microphonic (CM). This model of CM reveals that to match experimental data 1) stereocilia motion must be more sharply tuned than measured intra-OCC displacement responses, 2) the displacement-current gain of the mechano-electric transducer channels *in vivo* must be larger than what is measured *in vitro* by a factor of ~ 6 , and 3) the hair cells at more basal locations of the cochlea must be compromised. These predictions offer insight into aspects of cochlear mechanics that are not easily probed using OCT.

Table of Contents

Acknowledgments	xxii
Dedication	xxiii
Chapter 1: Principles of Spectral-Domain Optical Coherence Tomography	1
1.1 Architecture	1
1.2 Formation of an A-Scan	3
1.2.1 The OCT Signal in the Wavenumber Domain	3
1.2.2 The A-Scan	6
1.3 B-Scans, C-Scans and M-Scans	9
1.4 Spectral Domain Phase Microscopy	10
1.5 Resolution and Field of View	12
1.6 Practical Implementation	13
Chapter 2: Introduction to Cochlear Mechanics	14
2.1 Anatomy	14
2.2 Physiology	17
2.2.1 Tonotopy and the Traveling Wave	17
2.2.2 Nonlinearity and Zwuis Stimuli	21
2.2.3 Displacement Characteristics within the Organ of Corti Complex	26

Chapter 3: Resolving Anatomical Discrepancies in Uniaxial Recordings	28
3.1 Limits of Uniaxial Recordings	28
3.2 Illustration of Skew	30
3.3 Challenges in Interpreting Skewed Motion Measurements	33
3.4 Methods	34
3.4.1 Locally Planar Model	34
3.4.2 Determining the Approximate Anatomical Plane	36
3.4.3 Determining the Coordinate Transformation	40
3.4.4 Form of the Projection	41
3.4.5 Applications of Exploring Anatomical Space in Optical Coordinates	42
3.4.6 Implementation	43
3.5 Results	45
3.5.1 Quantification of Skew in the Gerbil Base	45
3.5.2 Skew Correction at Experiment Time	46
3.6 Validation	49
3.6.1 Radial Variation of Basilar Membrane Displacement	49
3.6.2 Sensitivity to Point Selection	50
3.7 Discussion and Conclusions	54
Chapter 4: Reconstructing Anatomical Components of Motion in the Organ of Corti Complex	56
4.1 Introduction	56
4.1.1 Challenges of Measuring Individual Motion Components	56
4.1.2 Challenges of Registration in Cochlear Mechanics	57

4.2	Theory of Reconstruction	59
4.2.1	The General Case	60
4.2.2	The Two-Dimensional Case	61
4.2.3	Error in the Two-Dimensional Case	62
4.3	Tonotopic Registration	64
4.4	Measured Longitudinal-Transverse Reconstructions	71
4.4.1	Explanatory Example	71
4.4.2	Longitudinal and Transverse Reconstructions at 70 and 80 dB SPL	73
4.4.3	Validation of Assumptions	75
4.5	Discussion	76
4.5.1	Comparison to Uniaxial Measurements	76
4.5.2	Physiological Implications of Broadband Nonlinearity	78
4.5.3	Lineal Motion	79
4.5.4	Potential for Troughs and Phase Lifts in Uniaxial Measurements	80
4.5.5	Challenges	84
Chapter 5:	Compressed Sensing Vibrometry	86
5.1	The Compressed Sensing Framework	88
5.1.1	Motivation	88
5.1.2	Form of the Data	89
5.1.3	Observation Model and the Form of the Objective Functions	91
5.1.4	Sparsity in a Wavelet Domain	92
5.1.5	Total Variation	94

5.1.6	Total Generalized Variation	95
5.2	Methods	97
5.2.1	Data Set Details	97
5.2.2	Pre-Processing	97
5.2.3	Evaluation	98
5.2.4	Implementation Details	98
5.2.5	Visualization of Areal Motion Maps	98
5.3	Results	99
5.3.1	Comparison of Methods	99
5.3.2	Uniform TGV Performance	102
5.4	Discussion and Conclusions	103
5.4.1	Superiority of Uniform Subsampling	103
5.4.2	Interpretation as a Denoiser	105
5.4.3	Experimental Applications	106
Chapter 6: Model of the Cochlear Microphonic Exploring Tuning and Magnitude of Hair Cell Transduction Current		
6.1	Introduction	108
6.1.1	Challenges in Measuring Stereocilia Motion Using OCT	108
6.1.2	The Cochlear Microphonic	109
6.1.3	Interpreting and Modeling the Cochlear Microphonic	111
6.2	Features of Cochlear Microphonic Data	112
6.3	Methods	114
6.3.1	Basics of the Model	114

6.3.2	The Boundary Value Problem and Physical Parameters	116
6.3.3	Implementation	117
6.3.4	Current Source	117
6.4	Results for Input Based on BM Displacement	120
6.4.1	Modeled Voltage Response	120
6.4.2	Comparison to Experimental Data	121
6.4.3	Premature Plateau	124
6.5	Results for Input with Enhanced Tuning	125
6.5.1	Comparison to Experimental Data	127
6.6	Quantitative Difference in Sensitivity	127
6.7	Outer Wall Conductivity	129
6.7.1	Effects of Varying the Wall Conductivity Ratio	129
6.7.2	Choice of the Outer Wall Conductivity Ratio	131
6.7.3	Relationship to Phase Accumulation	131
6.8	Causes of the Phase Plateau and Notch Behavior	132
6.8.1	Expected Relationship to the Phase Plateau and Notches	132
6.8.2	Effects of Damage	132
6.8.3	Relationships between Notch Frequencies	134
6.8.4	Furosemide Recovery Results Indicating Effects of Basal Damage	136
6.9	Conclusions	139
	Conclusion	141
	References	144

Appendix A: Experimental Details	153
A.1 Animal Preparation	153
A.2 Data Acquisition	154
A.3 Software Details	155
Appendix B: Wentzel-Kramers-Brillouin Method	156
Appendix C: Derivations of Algorithms for Compressed Sensing Vibrometry	158
C.1 Convex Optimization	158
C.1.1 Convex Functions and Conjugates	158
C.1.2 Primal-Dual Saddle Point Problems	159
C.1.3 Dual Norms	160
C.1.4 Proximal Gradient Descent and Ascent	161
C.2 Algorithms	164
C.2.1 Handling Complex Data	164
C.2.2 ISTA: Wavelet-Domain Sparsity	165
C.2.3 Total Variation	166
C.2.4 Total Generalized Variation	168
Appendix D: Wavelet-Domain Sparsity of Areal Displacement Maps	171
D.1 Sparsity in Wavelet Domains	171
D.2 Difference between Levels	171
Appendix E: Variations of the Finite Element Model	174
E.1 Longitudinal Variations	174

E.1.1	Effect of Longitudinal Variation in Scala Tympani Radius	174
E.1.2	Effect of Longitudinal Variation of Transducer Gain	176
E.2	Effects of Notches in the Current Source	177
	Publications and Conference Proceedings	180

List of Figures

- 1.1 Cartoon illustration of the architecture of a spectral domain OCT device, resembling a Michelson-Morley interferometer. A near-infrared source with wavenumber-domain (k -domain) spectrum $s(k)$ illuminates a beam splitter (the central white rectangle), creating the reference beam E_R and the sample beam E_S . Optical path length from the beam splitter is z . The galvo mirror directs the sample beam. The reflected beams interfere and are passed through a diffraction grating, after which their intensity is measured by a line camera. 2

- 1.2 Simulated example illustrating the acquisition of $\tilde{A}$ for a simple sample containing two point-reflectors. **A** – Arrow represents the sample beam travelling through a sample consisting of two reflective planes. The uppermost black curve represents the wavenumber-domain spectrum of the incident light (the background). Blue and orange curves correspond to the wavenumber-domain interference patterns induced by the two point-reflectors. Overlaid distances represent values of $z - z_R$ at the point-reflectors. **B** – Wavenumber-domain OCT signal, which is the sum of the black, blue and orange curves in **A**. **C** – Magnitude of the Fourier transform of the OCT signal in **B**, plotted against $z - z_R$. A large DC component comes from the background spectrum (black curve in **A**), while the marked blue and orange peaks correspond to the respectively colored reflectors. 8

- 2.1 Four images of the cochlea at different scales and perspectives. **A** – Cartoon of the cochlear spiral showing the gross shape and the direction of fluid pressure procession. The stapes of the middle ear pushes on the oval window (OW), and fluid volume is conserved by an opposite displacement at the round window (RW). A cross-section is shown revealing the three fluid compartments. The basilar membrane (BM), scala tympani (ST), tectorial membrane (TM) and auditory nerve are labeled, and shown in more detail in the other panels. **B** – Midmodiolar cross-section of the guinea pig cochlea from Slepecky, with the base at the bottom and the apex on top [10]. Labels: scala tympani (ST), scala media (SM), scala vestibuli (SV), modiolus (Mo), basilar membrane (BM), Reissner’s membrane (RM), stria vascularis (St.V), spiral ligament (Sp.L). **C** – Cartoon longitudinal cross-section of the organ of Corti complex at the base. The anatomical coordinate axes at this position, longitudinal (*L*), radial (*R*) and transverse (*T*), are shown at the top-right. Labels: round window membrane (RWM), pillar cells (PC), basilar membrane (BM), inner hair cells (IHC), outer hair cells (OHC), tectorial membrane (TM), Reissner’s membrane (RM). **D** – Scanning electron micrograph of a radial cross-section of the outer hair cells (OHCs) and Deiters cells (DC) in mole rat, from Raphael *et al* [11]. This cross-section is an orthogonal plane to the cartoon in **C**, as can be seen from the coordinate axes in the bottom-left. The phalangeal processes can be seen proceeding from the OHC-DC junction to the reticular lamina (RL) in the base-to-apex direction. The stereocilia of the OHCs can be seen on the RL, as the TM has been removed in the imaged preparation. 15
- 2.2 Snapshot of the traveling wave in response to a 1 kHz tone at a single instance in time. The blue curve is transverse velocity of the BM normalized by the response’s maximum value over the length of the cochlea. Dashed red curves indicate the envelope of the response, or the maximum magnitude that the wave will reach across time. Features to note are the peaking of the response at a best place and the spatial variation of the wavelength (or reciprocally, the wavenumber). A Wentzel-Kramers-Brillouin (WKB) approximate solution to a box model of the cochlea was used to generate these curves (see App B for more details). 18
- 2.3 Normalized velocity responses showing the two reciprocal tonotopy phenomena apparent in cochlear mechanics. A Wentzel-Kramers-Brillouin (WKB) approximate solution to a box model of the cochlea was used to generate these curves (see App B for more details). **A, B** – Velocity magnitude and phase responses along the longitudinal axis in response to single-tone stimuli at 1.0 kHz and 3.2 kHz. Responses show that frequencies are tuned to specific locations. Values are normalized so that the maximum velocity is 1. **C, D** – Frequency responses at two longitudinal positions, showing that each position is tuned to a specific frequency. The more apical position is 18 mm from the base and the more basal position is 9 mm from the base. 20

2.4	Illustration of compressive nonlinearity in mechano-electric transduction (MET) using a Boltzmann function model. A – A cartoon of an OHC. Deflection of the stereocilia results in a variation in the current through the cell. B – A graph of a sample Boltzmann function relating stereocilia deflection and MET current. The maximum current, linear range and operating point have been set to 1.5 nA, 75 nm and -25 nm respectively. Parameters were chosen based on <i>in vitro</i> recordings of OHC MET current [37]. Dotted lines indicate the operating point. C – A sample time-domain stereocilia deflection signal, acting as an input to the MET channel. Its frequency is 1 kHz and its peak-to-peak amplitude is 200 nm. D – MET current output for the input in C , according to the Boltzmann relationship in B . The dotted line indicates the MET current when the displacement is 0. Note that the signal saturates at both the minimum and maximum current values, but more-so at the minimum due to the negative operating point.	22
2.5	Displacement and gain magnitude responses from the gerbil base <i>in vivo</i> (Ge988), illustrating the manifestation of compressive nonlinearity. A , B – Displacement and gain at the BM in response to a multitone Zwuis stimulus containing 25 frequency components at 60, 70 and 80 dB SPL. C , D – The same, but at the OHC-DC junction.	24
2.6	Phase responses from BM and OHC-DC in the gerbil base <i>in vivo</i> . A – Responses from Ge961, measured near the 50 kHz region. OHC-DC lags BM across frequency, with this lag increasing with frequency. B – Responses from Ge967, measured near the 26 kHz region. OHC-DC leads BM across frequency, with this lead varying non-monotonically with frequency.	27
3.1	Anatomical and optical cross-sections in the gerbil base (~25 kHz region), resembling one another. A – Cartoon of the gerbil cochlea with the base at the top and apex at the bottom. B-Scans are taken through the RWM, capturing a basal cross-section of the OCC. B – Anatomical drawing of an <i>rt</i> -cross-section of the gerbil OCC with key anatomical structures labeled. The drawing is oriented as we would expect the OCC to appear in a B-Scan according to panel A . C – A B-Scan taken through the gerbil RWM as shown in the cartoon of panel A . B-Scans are inherently label-free, so anatomical structures are not immediately recognizable. D – The B-Scan from C with key anatomical structures labeled and color-coded. These labels are determined by comparison of the B-Scan in C with the anatomical drawing in B . ST = Scala tympani, SM = Scala media, SV = Scala vestibuli, HC = Hair cell, PC = Pillar cells, DC = Deiters cells, ISS = Inner sulcus space, OT = outer tunnel, RM = Reissner's Membrane. This figure is adapted from Fig 1 of <i>Frost et al., 2023</i> [49].	31

- 3.2 Two orthogonal views from an OCT volume taken through the gerbil RWM, revealing the difference between anatomical and optical coordinates. **A** – Cartoon of an anatomical rt -cross-section from Fig 3.1. **B** – Optical yz -cross-section with BM (blue) and OHC-DC (yellow) marked according to the known anatomy of **A**. **C** – Cartoon of an anatomical lt -cross-section through the OHC and DC, tilted. **D** – Optical xz -cross-section from the same volume as **B**, with BM and OHC labeled according to the anatomical cartoon in **C**. Together, these B-Scans show that the optical coordinate axes are distinct from the anatomical coordinate axes. 32
- 3.3 Illustration of optical, anatomical and approximate anatomical coordinates. **A** – B-Scan taken through the RWM of gerbil (same as in Fig 3.1) with optical coordinates labeled. **B** – Cartoon of the cochlea with the anatomical coordinate frame displayed as a single location. The frame spirals about the cochlea. **C** – Cartoon of a small region of the BM as a flat, spiraling sheet. The approximate plane is shown in gray, with the approximate anatomical coordinates labeled. These coordinates are static relative to Cartesian space. It can be seen that the planar approximation only appears reasonable over a small region of the BM, degrading as one moves further along the BM spiral. The dashed line indicates where the orienting B-Scan from **A** would lie. The labeled points **A**, **B** and **C** are used to determine the approximating plane according to the process described in Sec 3.4.2. This figure is adapted from Fig 2 of *Frost et al., 2022* [18]. 35
- 3.4 Analysis of the validity of the planar approximation in the gerbil base. The ten black circles indicate points selected along the medial edge of the spiraling BM over a $440\ \mu\text{m}$ longitudinal span. Each panel shows the projection of the selected points onto a different optical plane, along with a projection of the best-fit line. Black lines show the planar approximation over half of the span ($220\ \mu\text{m}$), while gray lines show the planar approximation over the full span. R^2 values are shown in each panel for both ranges. This figure is adapted from Fig 9 of *Frost et al., 2023* [49]. 37
- 3.5 Example of the plane approximation process using a volume scan from Ge900 (25 kHz region, taken through RWM *in vivo*). Two B-Scans from a single volume scan, $20\ \mu\text{m}$ apart. Points **A** and **B** are chosen in the first B-Scan (left). This determines the line segment that approximates the BM in this cross-section, shown to the user for verification before moving to the next step. In the second B-Scan (right), **C** is chosen, completely defining the plane. The projection of that plane onto the second B-Scan is shown to the user for verification before the values describing the plane are saved. This figure is adapted from Fig 5 of *Frost et al., 2022* [18]. 39

3.6	A still of the orienting GUI used to explore volumes in both optical and anatomical coordinates. Shown is the orienting B-Scan from experiment 900. The coordinate values of the blue point's location are shown in the anatomical (left) and optical (right) systems, and the A-Scan in which this point lies is shown in white. The red line is the planar approximation of the BM projected onto the displayed B-Scan. This figure is adapted from Fig 3 of <i>Frost et al., 2022</i> [18].	44
3.7	Illustration of longitudinal skew in Ge900. The panels in the top row shows the same B-Scan and A-Scan (white line), with three measured points isolated. The first is on the BM, to which other measurements are referenced. Cartoons in the bottom row show the anatomical structures at which the measured points are expected to lie, and are labeled by their longitudinal distance from BM. In particular, measured OHC-DC lie $46\ \mu\text{m}$ apical of BM.	46
3.8	Use of the orienting GUI to measure BM and OHC in the same anatomical cross-section. A – To begin, we select an A-Scan containing BM and OHC. The zero point is set to be on the BM along this A-Scan, as shown here. B – We move the z slider so that the blue point is on the OHCs and the A-Scan has not changed. Only the z optical coordinate changes, whereas all three anatomical coordinates have changed. The l value indicates that the OHCs are about $45\ \mu\text{m}$ apical of the BM in this A-Scan. C – We find the measurement location necessary to measure BM motion in the same cross-section as OHC from the previous A-Scan by moving to the point with the same l position but with $r = t = 0$. The OCT (X, Y) coordinates on the bottom right are the output we use to direct the OCT scanner to the desired measurement location. D – We display the measured displacement phase with respect to ear canal (EC) at the OHC and BM in the first A-Scan, as well as the BM in the second A-Scan. The OHC from run 1 and BM from run 2 are in the same anatomical cross-section. Data taken at 80 dB SPL. This figure is adapted from Fig 6 of <i>Frost et al., 2022</i> [18].	47
3.9	Illustration of the method used to test the operation of the orienting GUI against known physiology. The blue band in each view indicates the BM. The magenta dots in the expanded view on the right indicate points that span the BM radially. The white box is the xy -plane of a volume scan, with the interior white lines indicating the y -axis of the B-Scans in which the magenta points lie. The orienting GUI identifies the locations of the magenta points, all of which lie in different B-Scans. This figure is adapted from Fig 7 of <i>Frost et al., 2022</i> [18].	49

3.10	BM displacement data from Ge900 at two different head angles provide evidence that our method correctly identified the anatomical radial axis. A and B – BM displacement amplitude and phase taken along a single anatomical cross-section at seven radial locations spaced $10\ \mu\text{m}$ apart medial (aqua) to lateral (yellow), with locations approximated in C . D and E – BM displacement amplitude and phase taken in a different anatomical cross-section at radial locations spaced 20 or $40\ \mu\text{m}$ apart medial (aqua) to lateral (brown), with locations approximated in F . Data taken at 80 dB SPL. Figure adapted from Fig 8 of <i>Frost et al., 2022</i> [18].	51
3.11	Angular and distance error induced by perturbation in selected points, as defined in Eqns 3.11 and Eqn 3.12. A – B-Scan with a $40\ \mu\text{m} \times 40\ \mu\text{m}$ box centered around a selected point A , representing a range of possible selected points for which error will be computed. B – error induced by selecting points within this box, holding points B and C fixed. C , D – Same as A and B , except varying point B while holding A and C fixed. Fig 4 from <i>Frost et al., 2022</i> [18].	52
4.1	Sample B-Scans from Ge995 displaying the reflection property of OCT images. A-B – Anatomical cartoon and corresponding B-Scan showing the RWM “right-side-up”, appearing nearer to the scanner than the OCC. C-D – Similar to A-B , except that the scanning head has been zoomed towards the sample. The RWM now appears reflected as if it were further from the scanner than the OCC.	58
4.2	Graphical representation of the steps followed in the experimental method employed for two-dimensional reconstruction. A – Labeled B-Scans from a single volume, with two points p ₁ and p ₂ at the same anatomical structure (OHC-DC) marked in two different cross-sections. B – The BM approximated as a plane, in which the longitudinal direction connects any two points at the same anatomical structure between cross-sections. C – Cartoon of the BM with many measurements taken η apart longitudinally; anatomical axes with the measurement axis represented in longitudinal and transverse components. D – Cartoon of OCC with points labeled at BM and OHC-DC, along with an A-Scan with these same points labeled; Δ_i is the axial distance between OHC-DC and BM. E – Cartoon of the BM with OHC-DC and BM in the same anatomical cross-section but different A-Scans aligned by the known longitudinal component of the measurement axis. This figure is adapted from Fig 3 of <i>Frost et al., 2023</i> [49, 67].	65

- 4.3 Graphical representation of the method by which cross-sections and OHC-DC points are registered between orientations post-experiment. The pink and blue rectangles represent the approximately linear BM segments observed at the two orientations. **A** – On the left, the first measured BM point’s phase response at 80 dB SPL, $\phi_{1,1}(f)$; on the right, a set of BM responses at 80 dB SPL taken 10 μm apart longitudinally at the second orientation, $\phi_{2,n}(f)$ for $n = 1, 2, 3, 4, 5$. **B** – $\phi_{1,1}(f)$ and the nearest phase response at orientation 2, ϕ_{2,m_0} . The two phase responses are nearly identical, showing that the BM points measured at $b_{1,1}$ and b_{2,m_0} are in the same anatomical cross-section. **C** – Cartoons of the BM at both orientations with registered cross-sections shown by dotted lines. OHCs aligned to these BM points, $\mathbf{o}_{1,j}$ and $\mathbf{o}_{2,k}$, determined as in Fig 4.2 **E**, are thereby also in the registered anatomical cross-section. All four of these measured points lie in the same anatomical cross-section, and the OHCs are thereby registered to one another as marked. This figure is adapted from Fig 4 of *Frost et al., 2023* [49]. 66
- 4.4 Cartoon of a longitudinal-transverse cross-section of the OCC containing the DCs and OHCs. Measurement axes at two orientations that are achievable through the gerbil RW are shown. Measured OHC and BM positions along one measurement axis will lie at different longitudinal locations. Eleven measurements at each angle are taken 10 μm apart longitudinally. The boxed region shows where we can align measured BM positions with measured OHC-DC positions – this corresponds to about six distinct OHC-DC positions being registered. The arrow in the top-left shows the direction of motion in which the OHC-DC region was found to move in the reconstruction of Fig 4.5. This figure is adapted from Fig 5 of *Frost et al., 2023* [49]. 70
- 4.5 An example of responses and corresponding reconstruction from Ge967 at registered OHC-DC positions performed using responses to an 80 dB 15-frequency, 1 sec Zwuis stimulus. **A, B** – Magnitude and phase responses at aligned OHC-DC positions taken at two orientations – viewing angles 1 and 2 make 64° and 50° angles with the BM normal, respectively. BM responses in the registered cross-section measured at both orientations are also shown. Note that the BM phase responses are nearly indistinguishable between orientations, indicating that the cross-section is truly registered. Similarly, the BM magnitudes are parallel (offset vertically), and differ by a geometric factor determined by the ratio of the measurement angles’ cosines. **C, D** – Reconstructed longitudinal and transverse magnitude and phase responses at the OHC-DC, generated by application of Eq 4.7 to the data in panels **A** and **B**. For reference, we also show the BM phase response as a dashed black line, and the dashed gray line shows the reconstructed transverse phase shifted vertically by 0.5 cycles. **E** – DPOAE magnitudes in response to 70 dB SPL two-tone stimuli measured 20 minutes prior to the displacement measurements at each orientation. These two DPOAE measurements were taken one hour apart. This figure is adapted from Fig 6 of *Frost et al., 2023* [49]. 72

4.6	Transverse and longitudinal gain and phase responses reconstructed at the OHC-DC in response to Zwuis stimuli at 70 and 80 dB SPL at two distinct longitudinal locations. The measurement positions are spaced apart by $60\text{ }\mu\text{m}$ longitudinally. Light gray curves are reconstructed transverse OHC-DC responses, dark gray curves are reconstructed longitudinal OHC-DC responses and black dashed curves are BM responses in this same cross-section. This figure is adapted from Fig 8 of <i>Frost et al., 2023</i> [49].	74
4.7	Phase responses to 70 and 80 dB SPL Zwuis stimuli at OHC-DC (light gray) and BM (black). Comparison of phase reconstructed by our method to uniaxial measurements with strong longitudinal or strong transverse components. A, C – (Ge976) Uniaxially measured phase responses dominated by longitudinal motion in the gerbil base, $\text{BF} \approx 24\text{ kHz}$. Measurement axis made an $\sim 80^\circ$ angle with the BM normal. B, D – (Ge967) Reconstructed longitudinal phase responses, $\text{BF} \approx 26\text{ kHz}$. E, G – (Ge961) Uniaxially measured phase responses dominated by transverse motion in the gerbil base, $\text{BF} \approx 50\text{ kHz}$. Measurement axis made a $< 10^\circ$ angle with the BM normal. F, H – (Ge967) Reconstructed transverse phase responses from the same position as B and C , $\text{BF} \approx 26\text{ kHz}$. This figure is adapted from Fig 10 of <i>Frost et al., 2023</i> [49].	77
4.8	Gain (A) and phase (B) responses from Ge959, taken at a measurement angle containing significant transverse and longitudinal components. Of particular note are the amplitude trough and discontinuous phase lift seen in the 80 dB SPL response near 20 kHz.	81
4.9	Cartoon useful for visualizing the magnitude notch and phase discontinuity, provided by Dr. Elizabeth Olson. The OHC-DC motion line is shown (solid red top, solid light blue bottom), as well as its longitudinal-transverse components (dashed dark blue) and its components parallel and perpendicular to the measurement axis (dashed red top, dashed light blue bottom). The component parallel to the measurement axis is what is measured. In the top panel, longitudinal motion is of similar magnitude to transverse motion and our measurement axis is nearly perpendicular – only a small positive motion component is present along the beam axis. Without changing the measurement angle, a change in the longitudinal component magnitude yields a parallel motion in the opposite direction. If this shift occurs continuously in frequency, the beam axis will be perpendicular to the motion line at some point, at which phase will exhibit a discontinuous jump and magnitude will fall to 0.	83

- 5.1 Illustration of compressed sensing for image reconstruction. Losses are reported in normalized mean square error (NMSE) between the reconstruction and the original image. **A** – Image of Graham Elliot, America’s youngest four-star chef (**source**: IMDb). Inset on the bottom-left shows a zoom-in on his face, with his glasses being the feature of particular note. **B** – Subsampled version of the image in **A**, where a randomly chosen 50% of the original image coefficients have been set to 0. **C** – Image generated from the subsampled image in **B** using the total variation (TV) method discussed in this chapter (as well as in *Chambolle and Pock, 2016* [75]). Gross features look to be reconstructed well, while the inset shows weaknesses of the reconstruction for certain fine features. Normalized mean square error (NMSE): 1.6%. **D** – Same as **C**, except that the total generalized variation (TGV) method discussed in this chapter (as well as in *Chambolle and Pock, 2016* [75]) has been used for the image reconstruction. NMSE: 0.6%. 87
- 5.2 Sample vibration map with examples of subsampling patterns. **A** – Anatomical cartoon of the cochlea corresponding to the cross-section within which we measured motion. The measurement axis has strong transverse and longitudinal components, and the beam is swept across the longitudinal axis of the cochlea. **B** – Displacement magnitude map of Gerbil 988, 20 kHz component of the response to an 80 dB Zwuis stimulus. Colormap: 0 nm (black) to 6 nm (white). **C** – The same map as in **B** containing only 20% of the A-Scans, with subsampling occurring uniformly (i.e. gaps between samples are equal in width). **D** – The same map as in **B** containing 20% of the samples, with the subsampling pattern determined randomly (i.e. the width of each gap between samples is random). 90
- 5.3 Illustration of sparsity in a wavelet domain. **A** – One-channel image of Jerry C. LaPlante. **B-E** – A 1-level Daubechies-7 discrete wavelet transform (DWT) of the image in **A**; Approximation coefficients, horizontal detail coefficients, vertical detail coefficients and diagonal detail coefficients. **F** – Histogram of the approximation coefficient magnitudes. **G** – Histogram of the detail coefficient magnitudes (all three sets combined), showing that they are sparse as most of the coefficients have near-zero magnitude. 93
- 5.4 Normalized mean square error and magnitude SSIM for the three tested methods: TGV, TV and ISTA. Values displayed are sample means over a test set ($N = 20$), and error bars represent one sample standard deviation from the mean. Results are compared for both uniform and random subsampling by a factor of $P = 2, 5$ and 10. The inset in the NMSE panels shows the results for TGV on a smaller set of axes, as the error is far lower than those achieved using TV and ISTA. TGV with uniform subsampling is seen to significantly outperform all other methods – using this method at $P = 10$, the mean NMSE is less than 2%. 100

- 5.5 Example reconstructions using 10% of the M-Scans from the gerbil OCC. Colormap is shown in **A**, with hue representing phase re EC and saturation representing gain normalized to the maximum (further described in Sec 5.2.5). Sample: $\theta = 2$, 80 dB SPL, 10 kHz component. Maps are 100 rows by 200 columns, or $270 \mu\text{m} \times 300 \mu\text{m}$. Data are further described in Sec 5.2.1. Top Row: Results for uniform subsampling. **A** – densely sampled motion map. **B** – map from **A** subsampled uniformly by a factor of 10. **C** – dense map reconstructed using ISTA (2.38% NMSE). **D** – dense map reconstructed using TV (5.84% NMSE). **E** – dense map reconstructed using TGV (0.95% NMSE). Bottom Row: Results for random subsampling. **F** – densely sampled motion map (identical to **A**). **G** – map from **F** subsampled randomly by a factor of 10. **H** – dense map reconstructed using ISTA (45.55% NMSE). **I** – dense map reconstructed using TV (35.59% NMSE). **J** – dense map reconstructed using TGV (10.79% NMSE). Both qualitatively and quantitatively, TGV with uniform subsampling is seen to outperform the other methods on this sample. 101
- 5.6 NMSE and magnitude SSIM for TGV with uniform subsampling by factors of $P = 2, 5$ and 10 across the full dataset ($N = 275$), organized by orientation. At $\theta = 1$, $N = 75$; at $\theta = 2$, $N = 100$; at $\theta = 3$, $N = 100$. Values displayed are sample means over the set at each orientation, and error bars represent one sample standard deviation from the mean. Inset shows the NMSE for subsampling by a factor of 2, as it is much smaller than that achieved when subsampling by factors of 5 or 10. 103
- 5.7 Representative sample of three reconstructions made using TGV with uniform subsampling. Colormap is shown in **B**, with hue representing phase re EC and saturation representing gain normalized by maxima. **A** – Cartoon of approximate anatomy at $\theta = 1$ with basilar membrane (BM), Deiters cells (DC) and outer hair cells (OHCs) labeled. **B** – Densely sampled motion map for $\theta = 1$, 60 dB SPL, 15 kHz. Map is 100 rows by 200 columns, and $270 \mu\text{m} \times 300 \mu\text{m}$. **C** – Map in **B** uniformly subsampled by a factor of 10. **D** – Dense map reconstructed from the subsampled map in **B**. **E-H** – Same as **A-D**, but with $\theta = 2$, 80 dB SPL, 9 kHz. Maps are 100 rows by 200 columns, and $270 \mu\text{m} \times 300 \mu\text{m}$. **I-L** – Same as **A-D**, but with $\theta = 3$, 70 dB SPL, 25 kHz. Maps are 100 rows by 300 columns, and $270 \mu\text{m} \times 330 \mu\text{m}$ 104
- 6.1 Cochlear microphonic from experimental datasets. **A** and **B** – Set 1, Ge712 [3] amplitude and phase of CM measured close to ($\sim 20 \mu\text{m}$ from) the BM at the 16 kHz characteristic frequency (CF) location. SPL 20-90 dB in 10 dB intervals. **C** and **D** – Set 2, Ge693 [88] amplitude and phase of CM measured close to ($\sim 20 \mu\text{m}$ from) the BM at the 18 kHz CF location. SPL 30-80 dB in 10 dB intervals. **E** and **F** – Ge693, amplitude and phase of CM at various distances from the BM in scala tympani at the 18 kHz CF location, 45 dB SPL. Fig 3 of *Frost and Olson, 2021* [89]. 110

6.2	A – Cross-section of the gerbil cochlea, with the spiraling ST marked in blue. The red star represents the spiraling current source. B – Geometry of the model as it appears in the COMSOL Multiphysics user interface, representing an uncoiled version of the blue region in A . The outer wall is distinct from the larger fluid space, and the approximate position of the BM is marked by a half-cylindrical surface. The line current source can be seen on the flat surface. C shows a cross-section 2.5 mm from the base, and the vertical line from source to wall is where simulated voltages are recorded. OC = organ of Corti, BM = basilar membrane. Fig 1 of <i>Frost and Olson, 2021</i> [89].	115
6.3	The current source is initially assumed to be proportional to BM displacement shown here. A Amplitude and B phase of BM displacement, based on gerbil data with CF 15.5 kHz [98], at sound pressure levels 20-50 dB SPL. The phase was nearly independent of SPL and the small variations were not included. Phase is shown referenced to the ear canal pressure. The data are plotted versus frequency/CF. Inset in A shows enhanced tuning of hair bundle (HB) displacement input used in Sec 6.5. Fig 2 of <i>Frost and Olson, 2021</i> [89].	119
6.4	CM prediction under the assumption that current is proportional to BM displacement. $K = 50$, channel sensitivity = 33 pA/nm (starting value). Predictions are shown at five locations along the line segment 2.5 mm from the base of the cochlea (see Fig. 6.2 C). Magnitude and phase: A and B – at the position of the line current source; C and D – 55 μm from the source; E and F – 110 μm from the source; G and H – 160 μm from the source; I and J – 410 μm from the source. The dashed lines in the lower panels are the phase of the input (BM displacement) used to generate the current stimulus. Fig 4 of <i>Frost and Olson, 2021</i> [89].	121
6.5	Model LCM predictions 110 μm from the line current source ($\sim 20 \mu\text{m}$ from the BM) compared to experimental Set 1. Results (magnitude and phase) are shown at 20 (A and B), 30 (C and D), 40(E and F) and 50 dB SPL (G and H). CM predictions are based on the assumption that current is proportional to BM motion. $K = 50$, channel sensitivity is adjusted from starting value of 33 pA/nm to 200 pA/nm to align with the experimental result. The phase of the current stimulus is shown as a dashed line in the lower panels. Fig 5 of <i>Frost and Olson, 2021</i> [89].	122
6.6	Model LCM predictions 110 μm from the line current source ($\sim 20 \mu\text{m}$ from the BM) compared to experimental Set 2. Results (magnitude and phase) are shown at 30 (A and B), 40 (C and D) and 50 dB SPL (E and F). CM predictions are based on the assumption that current is proportional to BM motion. $K = 50$, channel sensitivity is adjusted from starting value of 33 pA/nm to 260 pA/nm to align with the experimental result. The phase of the current stimulus is shown as a dashed line in the lower panels. Fig 6 of <i>Frost and Olson, 2021</i> [89].	123

- 6.7 Model LCM predictions 110 μm from the line current source compared to experimental data. Current source is based on the enhanced tuning of HB motion. Comparisons are made at 40 dB SPL. **A** and **B** – Set 1 comparison. $K = 50$ and channel sensitivity adjusted from starting value to 200 pA/nm (same as Fig. 6.5). **C** and **D** – Set 2 comparison, $K = 50$ and channel sensitivity adjusted to 260 pA/nm (same as Fig. 6.6). The phase of the current stimulus is shown as a dashed line in each phase plot. Fig 7 of *Frost and Olson, 2021* [89]. 126
- 6.8 Effect of variations in lateral wall conductivity, $\sigma_w = \sigma/K$, where σ is the conductivity of the ST saline solution. Predicted CM at 20 dB SPL with $K=10, 25, 50, 100, 150$ and 300. **A** and **B** – 110 μm from the line current source; **C** and **D** – 210 μm from the line current source. Channel sensitivity is set to 33 pA/nm. Source is proportional to BM data if Fig 6.3. Fig 8 of *Frost and Olson, 2021* [89]. 130
- 6.9 Effect of nulling or reducing the current source basal or apical of the measurement location (19.5 kHz CF place). Panel sets show magnitude and phase at various distances from the current source . The original current source, based on BM displacement tuning, and the original channel sensitivity, 33 pA/nm, are used. In the nulled-base case, the current source from the base to the 21 kHz place is set to 0. In the reduced-base case the current source from the base to the 21 kHz place is reduced by half. In the nulled-apex case, current from the apex to the 18 kHz place is set to 0. SPL = 20 dB SPL, $K = 50$. **A** and **B** – CM predictions at the position of the line-current source. **C** and **D** – 55 μm from the line current source. **E** and **F** – 110 μm from the line current source. **G** and **H** – 160 μm from the line current source. **I** and **J** – 410 μm from the line current source. The phase of the current stimulus is shown as a dashed line in the lower panels. Fig 9 of *Frost and Olson, 2021* [89]. 133
- 6.10 Exploration of the basis of prominent notches. Amplitude and phase of the basilar membrane displacement data used to generate the model input. As in Fig. 6.3, except here the reference is stapes (so that all phase variation occurs within the cochlea). Highlighted in orange are the values of the amplitude and phase at the frequencies where notches appear in our model predictions, $\sim 0.8\text{CF}$ and 1.2CF . These correspond to phases of 0.36 and 2.36 cycles. Highlighted in blue are values corresponding to frequencies where the phases are half of a cycle off from the phase at the notch frequencies. Current components at the frequencies in blue will interfere destructively with those in orange. Fig 10 of *Frost and Olson, 2021* [89]. . 135

6.11	Experimental data related to model prediction with basal current nulled or reduced. CM measured close to the BM, before administering furosemide and after recovery from furosemide (3.5 hrs later). A – Normalized voltage amplitude. B – Phase relative to ear canal. Note that traveling wave phase accumulation is present even at high SPLs after recovery, indicating that the response was more local after recovering from furosemide than it was before administration of furosemide. A reasonable explanation is that the more basal region had not recovered fully, reducing the interference of non-local basal current. Fig 11 of <i>Frost and Olson, 2021</i> [89].	137
6.12	Experimental data related to model prediction with basal current nulled or reduced. OCT-measured displacement gain at the BM in the hook region of the gerbil cochlea (Ge971) before and after administration of furosemide. A – Baseline measurement made before administration of furosemide, showing the nonlinearity in the BF region characteristic of a healthy cochlea. B,C – Displacement gain measured in the same region 30 and 100 minutes after administration of furosemide, respectively. Furosemide has abolished the nonlinearity in the BF region, and this nonlinearity does not recover in any sense over the course of the experiment. This indicates that the base does not recover, as predicted by DPOAE measurements in [41].	138
6.13	Experimental data Set 2 (40 dB SPL) and LCM predictions as the current source representation progressed. The current source was initially represented in the model as proportional to BM displacement and transducer gain estimated from <i>in vitro</i> findings (cyan), then with increased transducer sensitivity (light blue), then enhanced tuning (medium blue) and finally reduction of the current sources basal to the measurement location (dark blue), which produced a more realistic phase accumulation. The phase of the current source was not changed and is included in a dashed line. Fig 12 of <i>Frost and Olson, 2021</i> [89].	140
D.1	Compression rates of <i>in vivo</i> areal cochlear motion patterns at three stimulus levels, seven stimulus frequencies and two viewing angles. Curves show the proportion of wavelet coefficients needed to achieve 1% NMSE from the original motion map using Daubechies 1-20 wavelet transforms. A-C : Compression rates for $\theta = 1$ in response to 60, 70 and 80 dB SPL stimuli, respectively. D-F : Same as A-C but for motion maps acquired at $\theta = 2$	172
D.2	NMSE for ISTA on the test set ($N = 20$) using two-, three- and four-level (L) Daubechies-7 wavelet transforms at $P = 5$ and $P = 10$	173
E.1	Model geometry wherein ST radius varies longitudinally approximately according to the known anatomy of gerbil, as it appears in the COMSOL Multiphysics user interface. The fluid space (OC, BM and ST) is shown in light blue, while the wall is shown in gray.	175

- E.2 Effect of more accurate geometry. CM prediction using the current source that is based on BM displacement at five locations along the line segment 2.5 mm from the base of the cochlea (19.5 kHz place), using two model geometries: that from Fig E.1 in which the radius varies, and that from Fig 1 in the main text in which the radius is constant. The presented “fixed radius” results are the predictions from Fig 6.4 SPL = 20 dB SPL, $K = 50$, channel sensitivity = 33 pA/nm. Magnitude and phase at the following distances from the current source: **A** and **B** – at the position of the line current source; **C** and **D** – 55 μm ; **E** and **F** – 110 μm ; **G** and **H** – 160 μm ; **I** and **J** – 410 μm . The phase of the current source is shown as a dashed line in the lower panels. 176
- E.3 Effect of a sharp notch in the current source. The notch is added to the original current source based on BM displacement, channel sensitivity = 33 pA/nm, $K = 50$, sound level = 20 dB SPL. **A** and **B** – magnitude and phase of the inputs with and without the notch. **C** through **L** show CM magnitude and phase at various distances from the current source. **C** and **D** – at the position of the line-current source; **E** and **F** – 55 μm ; **G** and **H** – 110 μm ; **I** and **J** – 160 μm ; **K** and **L** – 410 μm . The dashed lines in the lower panels are the phase of the BM displacement used to generate the current stimulus without the notch (as in **B**). 178
- E.4 CM prediction where the current source possesses a broad notch and a phase ripple, inspired by the resonant TM model of Nankali et al, Fig 5 [88]. $K = 50$ and channel sensitivity = 33 pA/nm. **A** and **B** – magnitude and phase of the current source. **C** through **L** show CM magnitude and phase at various distances from the current source. **C** and **D** – at the line-current source; **E** and **F** – 55 μm ; **G** and **H** – 110 μm ; **I** and **J** – 160 μm ; **K** and **L** – 410 μm . The dashed lines in the lower panels are the phase of the BM displacement used to generate the current stimulus (as in **B**). . . . 179

Acknowledgements

I would like to thank Drs. Elizabeth Olson and Christine Hendon for offering superb mentorship. Their knowledge, passion and openness have been invaluable to my growth as a scientist.

I would like to thank Dr. Stanislav Mintchev of the Cooper Union for training me in mathematics, and investing so much time and effort in my pursuit of academic research.

I would like to thank Dr. Clark Elliott Strimbu for the major role he has played in every experiment I've conducted in my time at Columbia. He has also been an excellent source of knowledge, and a pleasure to work alongside. I would also like to thank Dr. Erika Fallah and Lauren Chiriboga for being excellent labmates and friends.

I would like to thank scientists in the field of cochlear mechanics who have helped to edit and assess my work, and have taught me very much about the inner ear: Dr. Marcel van der Heijden, Dr. Jong-Hoon Nam, Dr. Karl Grosh, Dr. Sunil Puria and Dr. Tianying Ren.

I would like to thank Nikola Janjušević for pen-and-paper brainstorming sessions over weekly breakfasts, collaboration on the compressed sensing work, and mutual maintenance of sanity.

I would like to thank my parents, Kimberly Frost and Diana Butera, my brother Lance Weisberg, and my grandmothers, Claudia Sancimino and Diane Poland, for their unconditional love.

I would like to thank my partner Tosha Diamond-Huey, for sticking by my side for the last three incredible years.

I would finally like to thank my most beloved friends Michael Cohen, Mark Pirouet, Romaniya Voloshchuk, Hugo Ball, Karol Wadolowski, Dallas Skau, Shannon Jordan, David Benanti, Megan Noga, Dylan Phelan and Jaka Wescott for their continuous and immeasurable support.

Dedication

To Jerry C. LaPlante (1936-2020), incredible inventor and grandfather who sparked my love for science and engineering.

Chapter 1: Principles of Spectral-Domain Optical Coherence Tomography

Optical coherence tomography (OCT) fills a strong niche in medical imaging – it is capable of volumetric imaging several millimeters into a sample with typical resolutions on the scale of several microns in all three dimensions [1]. Additionally, it is capable of measuring sub-pixel displacements at the angstrom scale through spectral domain phase microscopy (SDPM) [2]. These features make OCT particularly well-suited for the study of cochlear mechanics, which concerns the complex sound-evoked vibration patterns of the organ of Corti complex (OCC) that are in part responsible for the sensation of hearing [3, 4, 5, 6, 7, 8].

OCT systems are generally classified into three categories: time-domain OCT, spectral-domain OCT and swept-source OCT. All data presented in this thesis was recorded using a ThorLabs Telesto 320 spectral-domain OCT system at Columbia University’s Fowler Memorial Laboratory (details in App A.2). As such, all future references to OCT refer to spectral-domain OCT in particular. In this chapter, I will discuss the mechanisms by which OCT can be used to acquire images, volumes and time-domain displacement signals.

1.1 Architecture

Fig 1.1 shows a diagram of the canonical spectral-domain OCT device architecture, which resembles a Michelson-Morley interferometer. Light from a broadband near-infrared source is split into two paths – the *sample beam* that travels towards the sample to be imaged, and the *reference beam* that travels towards a reference mirror. Before reaching the sample, the sample beam is redirected by a two-axis galvo mirror which precisely controls the angle of the beam. The sample beam reflects off the sample at various depths, and returns to the system to interfere with the reference beam.

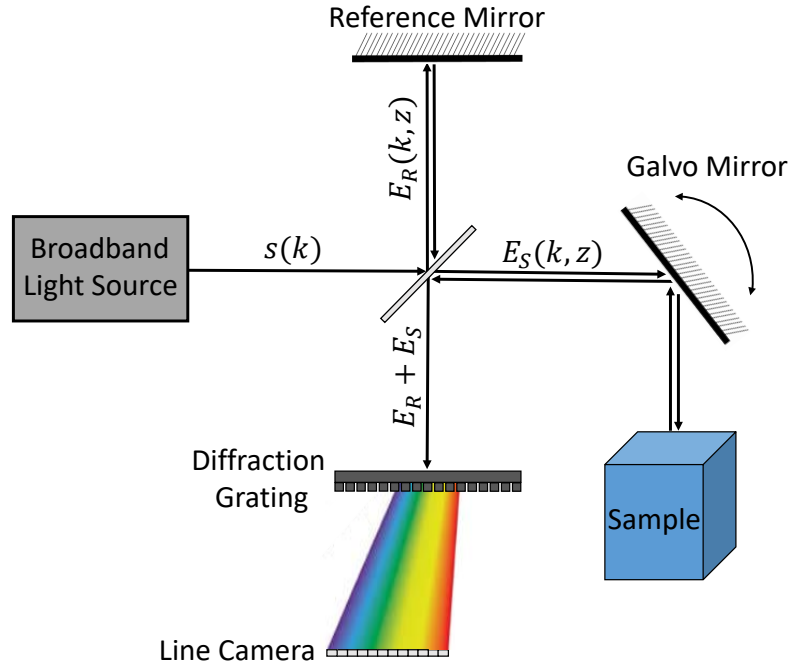


Figure 1.1: Cartoon illustration of the architecture of a spectral domain OCT device, resembling a Michelson-Morley interferometer. A near-infrared source with wavenumber-domain (k -domain) spectrum $s(k)$ illuminates a beam splitter (the central white rectangle), creating the reference beam E_R and the sample beam E_S . Optical path length from the beam splitter is z . The galvo mirror directs the sample beam. The reflected beams interfere and are passed through a diffraction grating, after which their intensity is measured by a line camera.

The interfering sample and reference beams pass through a diffraction grating, which spatially separates the light by wavelength. This illuminates a line camera for which each pixel corresponds to a different wavelength interval. This wavelength-domain intensity at the line camera is referred to as *the OCT signal*.

1.2 Formation of an A-Scan

The OCT signal is an interference pattern in the wavelength domain that contains the information required to generate a map of the reflectivity as a function of depth into the sample. This reflectivity map, known as an *A-Scan*, is the fundamental unit from which OCT images and displacement measurements are built. In this section, I will discuss the formation of an A-Scan from the OCT signal.

1.2.1 The OCT Signal in the Wavenumber Domain

For reasons that will soon become evident, I will represent quantities as functions of wavenumber k rather than wavelength λ , related by $k = 2\pi/\lambda$. I will also ignore noise phenomena for the time being.

Light is modeled as a plane wave traveling in the z direction, and is represented through its electric field. That is, each beam has the form

$$E(k, z, t) = A(k)e^{-j(kz - \omega t)},$$

where A [N/C] is the magnitude spectrum, z denotes the optical path length (OPL) traveled from the beam-splitter at $z = 0$, and ω is the temporal radian frequency. As time-dependence will be identical across all beams, it will be left implicit for the remainder of this section.

The light source has electric field spectrum $s(k)$ [N/C]. The sample and reference beams are given by

$$E_S(k, z) = s(k)a_S e^{-jkz}, \quad E_R = s(k)a_R e^{-jkz}$$

respectively. Here, a_S and a_R (both unitless) represent the proportion of light intensity split to their respective paths, satisfying $a_S^2 + a_R^2 = 1$.

The reference beam arrives at the reference mirror after traveling a distance z_R , reflects, and then arrives back at the beam splitter to interfere with the sample beam. That is, it interferes after traveling a distance $2z_R$. This is both the physical path length and OPL as the reference beam is assumed to pass through air with index of refraction 1.

Ignoring the distance from this point to the spectrometer (which is identical between the two beams), the detected reference beam is thereby

$$E_R^D(k) = s(k)a_R r_R e^{-jk2z_R}, \quad (1.1)$$

where r_R (unitless) is the reflectivity constant of the reference mirror. Ideally, $r_R \approx 1$.

The sample beam reflects off of the sample at various depths. The sample is modeled as a sequence of M point-reflectors, each at physical depth z_m into the sample with reflectivity r_m (unitless) for $m = 1, 2, \dots, M$. The total sample reflectivity function is

$$r_S(z) = \sum_{m=1}^M r_m \delta(z - z_m) \quad (1.2)$$

where δ is the Kroenecker delta function. The portion of the sample beam that reflects off of reflector m travels a total OPL of $2nz_m$ before arriving back at the beam splitter to interfere with the reference beam, where n (unitless) is the index of refraction of the medium.

We make the assumption that $r_m \ll 1$, so that almost all of the energy incident on the first reflector is also incident at the M^{th} reflector. We also assume that no scattering occurs, and that no internal reflection occurs. The detected sample beam is then

$$E_S^D(k) = s(k)a_S \sum_{m=1}^M r_m e^{-jk2nz_m}. \quad (1.3)$$

The right-hand side of the above equation can be recognized as the convolution of the plane wave

with $r_S(2nz)$.

The photodetector converts time-averaged incident power to current. The irradiance of the superposition of these two beams is

$$\frac{1}{2}c\epsilon_0|E_R^D + E_S^D|^2 \quad [\text{W/m}^2],$$

where c [m/s] is the speed of light and ϵ_0 [C²/Nm²] is the permittivity of free space. The photodetector has some responsivity $\mathcal{R}$ [A/W] and some pixel area $\mathcal{A}$ [m²]. Defining

$$\rho = c\epsilon_0\mathcal{R}\mathcal{A},$$

the OCT signal, I^D [A], is

$$I^D(k) = \frac{\rho}{2} \langle |E_R^D(k) + E_S^D(k)|^2 \rangle = \frac{\rho}{2} \langle (E_R^D(k) + E_S^D(k))(E_R^D(k) + E_S^D(k))^* \rangle, \quad (1.4)$$

where $\langle \cdot \rangle$ denotes integration over the photodetector's integration time, and $\cdot^*$ denotes complex conjugation.

Defining the spectrum intensity $S(k) = \langle |s(k)e^{j\omega t}|^2 \rangle$, I can write the rather large expression for measured intensity:

$$I^D(k) = \frac{\rho}{2}S(k) \left(a_R r_R e^{-jk2z_R} + a_S \sum_{m=1}^M r_m e^{-jk2nz_m} \right) \left(a_R r_R e^{jk2z_R} + a_S \sum_{m=1}^M r_m e^{jk2nz_m} \right) \quad (1.5)$$

$$= \frac{\rho}{2}S(k) \left(a_R^2 r_R^2 + a_S^2 \sum_{m=1}^M r_m^2 \right) + \quad (1.6)$$

$$+ \frac{\rho}{2}S(k) a_R a_S r_R \sum_{m=1}^M r_m \left(e^{j2k(nz_m - z_R)} + e^{j2k(z_R - nz_m)} \right) + \quad (1.7)$$

$$+ \frac{\rho}{2}S(k) \sum_{l \neq m} r_l r_m \left(e^{j2kn(z_l - z_m)} + e^{j2kn(z_m - z_l)} \right). \quad (1.8)$$

Three components are apparent here – the z -independent *DC term* in line 1.6 is a constant scaling

of the spectrum $S(k)$, the *cross-correlation* terms in line 1.7 represent interference between the reference beam and the components of the sample beam reflected from each reflector, and the *autocorrelation terms* in line 1.8 represent pair-wise interference between components of the sample beam reflected at distinct reflectors.

Earlier I assumed that $r_R \approx 1$ and $r_m \ll 1$ for $m = 1, 2, \dots, M$. Considering any pair of reflectors l and m , these assumptions give $r_l r_m \ll r_R r_m$. This means that the autocorrelation terms are small compared to the DC and cross-correlation terms, and will be ignored for the remainder of the derivation.

Removing the autocorrelation terms and using Euler's formula,

$$I^D(k) \approx GS(k) + HS(k) \sum_{m=1}^M r_m \cos[2k(nz_m - z_R)], \quad (1.9)$$

$$G = \frac{\rho}{2} \left(a_R^2 r_R^2 + a_S^2 \sum_{m=1}^M r_m^2 \right), \quad H = \rho a_R a_S r_R.$$

This is the compact form of the OCT signal in the wavenumber domain. Note that by the assumptions concerning reflectances, $G > H$.

1.2.2 The A-Scan

From Eqn 1.9 it can be seen that the OCT signal encodes the depth of each reflector by frequency. The natural next step is to consider the Fourier transform of this signal, where the Fourier partner of k is denoted as ζ [m]. Defining $\mathcal{F}\{I^D\} = \tilde{A}$ and $\mathcal{F}\{S\} = B$ (referred to as the background), the Fourier transform is

$$\begin{aligned} \tilde{A}(\zeta) &= GB(\zeta) + \frac{H}{2} B(\zeta) * \sum_{m=1}^M r_m (\delta[\zeta - 2(nz_m - z_R)] + \delta[\zeta - 2(z_R - nz_m)]) \\ &= GB(\zeta) + \frac{H}{2} \sum_{m=1}^M r_m (B[\zeta - 2(nz_m - z_R)] + B[\zeta - 2(z_R - nz_m)]) \end{aligned} \quad (1.10)$$

where $*$ denotes convolution and δ denotes either the Dirac or Kroenecker delta function (in the idealized continuous case or the practical discrete case, respectively).

This signal is conjugate-symmetric in ζ (as it is the Fourier transform of a real signal) and contains two components – a scaled copy of the spectrum centered about $\zeta = 0$, and scaled copies of the background centered at different ζ positions corresponding to the depths of the reflectors. This is nearly the desired object – a spatial reflectivity profile achieved by taking a Fourier transform of the OCT signal.

This concept is illustrated for a simple two-point-reflector sample in Fig 1.2. The nearer reflector (blue) produces a lower-frequency wavenumber-domain interference pattern, while the further reflector (orange) produces a higher-frequency pattern. These add with the DC component to generate the OCT signal in panel **B**. Its Fourier transform magnitude in **C** shows broad peaks at the positions of the two reflectors symmetrically distributed about the origin.

However, there are two apparent problems – 1) G is large compared to H , so the background spectrum centered at 0 may dwarf nearby signals corresponding to reflectors (see the size of the DC peak in Fig 1.2 **C**), and 2) each reflector is smoothed in the ζ domain by the background, so the resolution of the system is limited by the properties of the broadband light source (see the breadth of the peaks in Fig 1.2 **C**). Point (2) corresponds to the interpretation of $B(\zeta)$ as a point-spread function for the OCT system. For these reasons, $\tilde{A}$ is not used as the A-Scan.

Instead, we measure the spectrum $S(k)$ by blocking the sample beam. The measured $S(k)$ can be used in two ways. First, by making an estimate for G (this process usually involves observing the average value of the OCT signal), one can *subtract* $GS(k)$ from the OCT signal $I^D(k)$. This removes the DC term.

Then, one can *divide* this signal by $S(k)$. In the spatial domain, this is equivalent to *deconvolving* with $B(\zeta)$. This process is effectively identical to the standard optical procedure of deblurring by deconvolving with the point-spread function. After performing this subtraction and division in

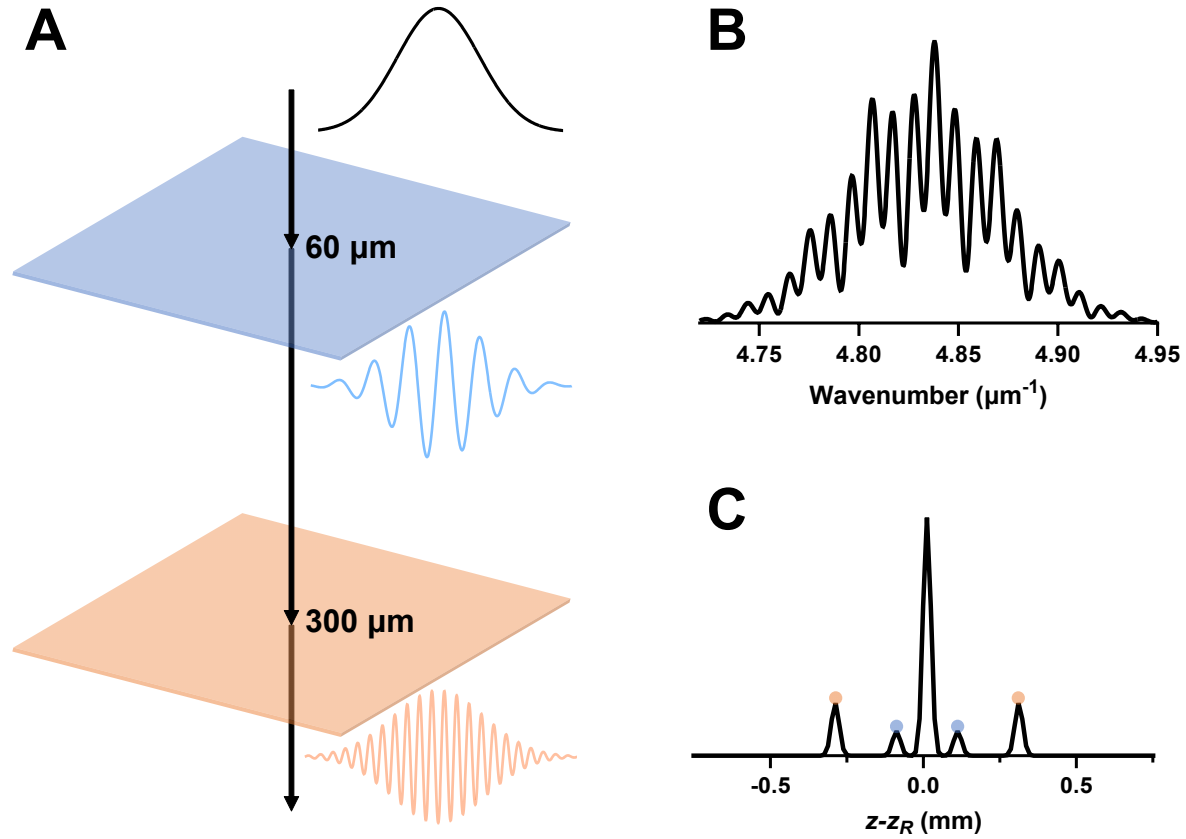


Figure 1.2: Simulated example illustrating the acquisition of $\tilde{A}$ for a simple sample containing two point-reflectors. **A** – Arrow represents the sample beam travelling through a sample consisting of two reflective planes. The uppermost black curve represents the wavenumber-domain spectrum of the incident light (the background). Blue and orange curves correspond to the wavenumber-domain interference patterns induced by the two point-reflectors. Overlain distances represent values of $z - z_R$ at the point-reflectors. **B** – Wavenumber-domain OCT signal, which is the sum of the black, blue and orange curves in **A**. **C** – Magnitude of the Fourier transform of the OCT signal in **B**, plotted against $z - z_R$. A large DC component comes from the background spectrum (black curve in **A**), while the marked blue and orange peaks correspond to the respectively colored reflectors.

the wavenumber domain, the A-Scan in the ζ domain is given by

$$\begin{aligned} A(\zeta) &= \mathcal{F} \left\{ \frac{GS(k) + HS(k) \sum_{m=1}^M r_m \cos[2k(nz_m - z_R)] - GS(k)}{S(k)} \right\}(\zeta) \\ &= \frac{H}{2} \sum_{m=1}^M r_m (\delta[\zeta - 2(nz_m - z_R)] + \delta[\zeta - 2(z_R - nz_m)]), \end{aligned} \quad (1.11)$$

Considering only positive ζ , assuming $nz_m > z_R$ for all reflectors, and letting $z = (\zeta + 2z_R)/2n$, we arrive at the formula for the ideal A-Scan in physical distance into the sample:

$$A(z) = \frac{H}{2} \sum_{m=1}^M r_m \delta[2n(z - z_m)]. \quad (1.12)$$

Normalizing the z axis by $2n$, we get the physical distance between the reflectors and the beam splitter.

1.3 B-Scans, C-Scans and M-Scans

The modulus of the A-Scan is a single-channel one-dimensional “image” along the beam’s axis, z , into the sample. To form a 2-D image, the sample beam can be swept along an axis y using the galvo mirror. At each point along the y axis, an A-Scan can be recorded to form what is known as a *B-Scan*. The modulus of a B-Scan is proportional to the reflectivity in a subset of the yz -plane, forming a grayscale image in the standard sense.

Naturally, the beam can also be swept along a third perpendicular axis x so that the corresponding unit vectors $(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}})$ form a right-handed coordinate system. Recording A-Scans while sweeping the beam across a rectangular area in the xy -plane, a volumetric reflectivity map known as a *C-Scan* (or equivalently *volume scan*) is formed.

Until now, time-dependence of the A-Scan has been left implicit. In the case of a stationary sample, the A-Scan is time-independent. However, in a non-stationary sample, the A-Scan may change significantly as a function of time. We refer to a series of A-Scans recorded at the same xy -location over a period of time as an *M-Scan*.

In the idealized case, the z axis is continuous. This means that motion is encoded simply by letting a reflector point z_m vary as a function of time. In practice, the spectrometer has only a finite number of pixels so that z is discretely sampled with pixel size Δz . The treatment of motion then falls into two categories – 1) when the displacements of reflectors varies on a scale larger than Δz , and 2) the case of *sub-pixel* displacements, where displacement magnitudes are smaller than Δz .

Displacements of type (1) are sufficiently large to be seen in the modulus of an M-Scan. This means that such displacements can be analyzed through standard video processing methods on the real M-Scan modulus signal.

Displacements of type (2) will be invisible in the modulus, as the reflectivity at each pixel does not change as a function of time. This case requires special treatment specific to the discretization of z -space. As the sound-evoked displacements in the OCC are on the order of nanometers – far less than the several-micron scale of the z -resolution of OCT – this case is of particular interest for the study of cochlear mechanics.

1.4 Spectral Domain Phase Microscopy

Consider a reflector with reflectivity r_0 and z -direction displacement $z_0 + d(t)$, where z_0 is the z -position of the nearest pixel boundary and $0 < d(t) < \Delta z$ (i.e. the reflector lies within a single pixel for all time). The OPL traveled by the portion of the sample beam reflected by this reflector is $2n(z_0 + d(t))$. The sample beam at the detector is

$$E_S^D(k, t) = s(k) a_S r_0 e^{-jk2n(z_0+d(t))}. \quad (1.13)$$

To determine the effect of this displacement on the OCT signal, one can associate this OPL with a single z_m in Eqn 1.9. Subtracting the background and dividing by S , as in Eqn 1.11, the ζ -domain A-Scan over time, or M-Scan, will be

$$M(\zeta, t) = \mathcal{F}_k \{ H r_0 \cos[2k(nz_0 - z_R) + 2knd(t)] \}(\zeta, t), \quad (1.14)$$

where the subscript k denotes that the Fourier transform is being taken with respect to wavenumber. As the time-dependent term is itself multiplied by k , its impact to the equation would modulate the detected reflector position – however, we know that is not a detectable phenomenon. We can circumnavigate this issue through a simple approximation.

Writing the central wavelength of the light source as λ_0 , we can approximate the k term multiplying $d(t)$ as a constant, $2\pi/\lambda_0$. That is, we can write

$$M(\zeta, t) \approx \mathcal{F}_k \left\{ H r_0 \cos \left[2k(nz_0 - z_R) + \frac{4\pi n d(t)}{\lambda_0} \right] \right\} (\zeta, t). \quad (1.15)$$

As the Fourier transform is being taken with respect to wavenumber, the time-dependent term acts as a k -independent phasing term. Considering only positive ζ , the Fourier transform is

$$M(\zeta, t) = \frac{H}{2} r_0 \delta[\zeta - 2(nz_0 - z_R)] e^{j \frac{4\pi n d(t)}{\lambda_0}}. \quad (1.16)$$

Under the same coordinate transformation as before, I have

$$M(z, t) = \frac{H}{2} r_0 \delta[2n(z - z_0)] e^{j \frac{4\pi n d(t)}{\lambda_0}}. \quad (1.17)$$

Referring to the phase of the M-Scan as $\angle M = \phi$, the phase of the measurement at z_0 is $\phi(z_0, t) = 4\pi n d(t)/\lambda_0$. This allows one to recover the sub-pixel displacement signal from the M-Scan as

$$d(t) = \frac{\lambda_0 \phi(z_0, t)}{4\pi n}. \quad (1.18)$$

This method is known as *spectral domain phase microscopy* (SDPM), and facilitates measurement of sub-pixel displacements with incredible angstrom-scale resolution [2]. It should be noted that this method only picks up the one-dimensional motion of the reflector along the z axis.

1.5 Resolution and Field of View

The resolution of the OCT system is determined by the central wavelength (λ_0) and bandwidth ($\Delta\lambda$) of the light source, as well as the numerical aperture of the objective lens (NA) [1]. The lateral (x and y) resolution is derived from the Rayleigh criterion, given in terms of the numerical aperture and central wavelength as

$$\delta x = \delta y \approx 0.37 \frac{\lambda_0}{NA}. \quad (1.19)$$

It should be noted that this formula is only valid at the focal plane of the system, so the lateral resolution is distinct at different z -locations in an OCT scan. This means that the objective lens also determines the axial (z) field of view.

The axial resolution of the system is determined by the coherence length, and can be shown to be

$$\delta z = \frac{2 \ln 2}{n\pi} \frac{\lambda_0^2}{\Delta\lambda}. \quad (1.20)$$

There exists a fundamental tradeoff between resolution and penetration depth. In particular, a larger central wavelength yields a poorer resolution, as can be seen in Eqn 1.20, but also tends to yield a larger penetration depth due to the phenomenon of Rayleigh scattering [1].

Axial field of view is also limited by *occlusion*. The A-Scan formula is valid under the assumption that the reflectivities within the sample are small. Of course, if the reflectivities were too small, almost no signal would be picked up by the device. On the other hand, if the reflectivities are too large, very little light is transmitted deeper into the sample. This is important for *in vivo* imaging of the cochlea, as the bony walls of the cochlea obscure views due to their high reflectivity. In this case, the practical axial field of view cannot extend beneath this opaque structure.

1.6 Practical Implementation

Certain details of the OCT data processing are omitted in the discussion above. In particular, the spectrometer data recorded in the wavelength domain must be interpolated to the wavenumber domain, and the signal must be windowed before the Fourier transform is applied.

At experiment time, data acquisition is controlled through a C++ program that uses ThorLabs' SpectralRadar SDK. This program uses built-in processing functions to perform the background acquisition, subtraction and normalization, as well as the interpolation, windowing and Fourier transform steps. The output of this program is a complex-valued time series of A-Scans. All other processing steps are performed after experimentation in MATLAB.

Data presented in this thesis were recorded using a ThorLabs Telesto 320 system. Details of this system are presented in App A.2.

Chapter 2: Introduction to Cochlear Mechanics

The sensation of hearing in mammals begins with a pressure wave entering the ear canal (EC). EC pressure generates a displacement at the tympanic membrane, which in turn pushes on the ossicles of the middle ear. The last of these bones, the stapes, pushes on the oval window (OW) of the cochlea and modulates the pressure of the fluid within [9].

The cochlea is then responsible for transforming this fluid pressure into a neuronal stimulus, and it does so in a fascinating manner. In this chapter, I will discuss the anatomy of the cochlea and properties of its sound-evoked displacement responses.

2.1 Anatomy

The bony exterior of the cochlea is in the shape of a snail's shell, with the tip of this shell being referred to as the *apex*, the broad end as the *base*, and the central axis of the spiral as the modiolus. Fig 2.1 **A** shows a cartoon of the cochlear spiral with the base on the top-left and the apex at the center. Fig 2.1 **B** shows a "midmodiolar" cross-section of the guinea pig cochlea, with the modiolus at the center, the base at the bottom and the apex at the top [10].

Three fluid ducts, or *scalae*, fill the cavity (see Fig 2.1 **B**). The scala vestibuli (SV) contains sodium-rich perilymph, and abuts the oval window (OW) at the base where the stapes footplate supplies the cochlea's input force. Opposite SV is the scala tympani (ST), also containing perilymph, which abuts the round window membrane (RWM) at the base [10]. SV and ST are connected at the most apical position in the cochlea known as the helicotrema. As fluid is pushed in and out at the OW, the perilymph volume is conserved by an opposite displacement at the RWM (illustrated by the dotted arrows in Fig 2.1 **A**)[12, 13].

Between SV and ST lies scala media (SM), filled with potassium-rich endolymph. SV and SM